

Immune-Mediated Polyarthritis (IMPA)

What is polyarthritis?

Poly (many) + **arthr** (joint) + **itis** (inflammation) = inflammation in **more than one joint**. Inflammation is the body's response to injury, infection or an overactive immune system.

Causes

- **Infections** (e.g., tick-borne diseases)
- **Reactive** — triggered by inflammation elsewhere (heart, uterus, bladder, skin, intestines)
- Drugs or vaccines; some cancers
- **Idiopathic** — no cause found; the immune system attacks the joints (the most common form)

Who is affected?

Usually dogs **3–7 years old** (range 6 months–12 years), any breed, males and females equally. Rare in cats.

Signs

- **Stiff, short-strided walking**; reluctance to stand or walk
- **Fever**, lethargy, poor appetite — often looks like a "sick all over" dog
- Wrists and ankles most often affected; joints may be swollen, but often are not
- May occur with inflammation of the brain, muscles, skin, liver or intestines

Diagnosis

- Physical exam; blood tests, urinalysis (checking for protein loss) and **blood pressure**
- **Tick-borne and other infectious disease testing**

- X-rays of the chest and joints; sometimes abdominal ultrasound
- **Joint taps** — fluid from at least **4 joints** is needed to diagnose IMPA. Many inflammatory cells suggest IMPA (or infection).

Treatment

- **Prednisone** daily until signs resolve, then gradually reduced
- A second immune-suppressing drug is often added so steroids can be lowered:
 - **Mycophenolate** — diarrhea is common; blood problems are rare
 - **Leflunomide** — mild side effects (poor appetite, tiredness, vomiting); regular blood tests needed
 - Other options are available
- If signs return, long-term medication may be needed

Follow-up

- If there is **no improvement within 7 days** of steroids, the diagnosis is re-evaluated
- Regular blood and urine tests; rising urine protein may signal kidney damage
- A blood test (CRP) may help track response; repeat joint taps are not usually needed

Outlook

About **90%** respond quickly to treatment. About 10% do not and develop ongoing lameness and joint damage. Long-term steroid side effects are a common concern, which is why we aim to taper to the lowest effective dose.